Meridian expansions and classical symbols of quantum recurrences

Abstract

For fixed strict Conway diagrams with independently long powers, and for circle trees satisfying a summed or directed degree–twist criterion, we prove meridian expansions to every inverse-level order. They hold on fixed translated microscopic charts and retain the full normal coordinates at repeated spectral roots. At order N^{-r} the logarithmic coefficient has degree at most $r+1$ in the microscopic parameter. Its highest degree is the stationary value of an explicit analytic deformation of the original source. We then prove that every specified fixed recurrence annihilating the whole reduced invariant at all sufficiently large levels has this analytic characteristic germ in its first nonzero classical symbol. The reduced-to-full gauge retains both its meridian divisor and its sign. A separate marked geometric identification determines when the germ is the preferred meridian–longitude branch.

1 Fixed diagrams, normalization and conclusions

Throughout, a diagram is one fixed literal ordered tangle expression, with its physical turns, original crossing circles, endpoint orders, cut and zero-framing convention. Its closure is one knot and no component is closed inside a proper tangle. All signed integer crossing powers n_v are fixed before N tends to infinity. We use one of the following already proved source classes:

1. a strict Conway expression with at least two original circle vertices and all $|n_v|$ above the single diagram-dependent threshold of [3, Theorem 1.2];
2. an all-circle tree with at least one edge, degrees d_v and $\sum_v d_v/(d_v^2+n_v^2) \le 2^{-25}$, as in [6, Theorems 1.1–1.2];
3. an all-circle tree with a root and numbers c_v satisfying [7, equations (2.3), (2.8), (2.9), Theorem 1.1].

The last is a finite criterion, not a condition on an unknown invariant. It includes, by [7, Corollary 1.2], the separate alternative $d_v \ge 2^{28}$ or $|n_v| \ge 2^{27}$ at every vertex, and the local condition $\max_v d_v/(d_v^2+n_v^2) \le 2^{-120}$. In these all-circle cases large degree can permit $n_v = 0$. No long-power assumption is added to them here. The explicit directed inequalities are recalled in Section 4; no summed hypothesis is substituted for that criterion.

Let $F_N(M)$ be the actual zero-framed reduced state of [2], with $q_N = e^{\pi \mathrm{i}/N}$ and marked meridian $M = q_N^s$. In its highest-weight coordinate the pointing is $M = -q_N x$; the source flag has eigenvalue M^{-1}. Put

$$K_N = F_N(1), \qquad H_N(s) = \frac{\sin \pi s}{N \sin(\pi s/N)}, \qquad \mathcal{Z}_N(s) = \frac{F_N(q_N^s)}{K_N H_N(s)}. \tag{1.1}$$

The value of H_N at zero is one. The imported source theorems prove $K_N \neq 0$ for all sufficiently large levels, in both parities. For $N \ge 3$ it is the reduced Kashaev value with the

fixed complete normalization. The one factor H_N is the actual reduced/full conversion, not an adjustable regularization.

The local action needs its full normalization. Let X collect all original physical wire coordinates x_e, all hidden unturned product coordinates where present, and the original circle angles τ_v. Define

$$\begin{aligned} J(x) &= \int_0^x \log(2\sin \pi u)\, du, \\ b(x,\tau) &= \int_0^x \log(2\cos 2\pi u + 2\cos\tau)\, du - 2J(x), \\ m(x,y,z) &= J(x+y) + J(z) - J(x) - J(y) - J(z-x) - J(z-y) - J(x+y-z). \end{aligned} \tag{1.2}$$

The logarithms continue the original positive complete branch. For strict Conway sources retain the complete bare gates and set

$$\Phi(X) = \sum_{\text{bare gates}} m(x,y,z) + \sum_{v \text{ circle}} \sum_{e \ni v} b(x_e, \tau_v) + \frac{\mathrm{i}}{4\pi} \sum_v n_v \tau_v^2 - \mathrm{i} \sum_v \tau_v. \tag{1.3}$$

For all-circle trees the first sum is absent. The last term is the original first Fourier charge and is part of the mark. Write X_0 for its selected filled critical point. The source theorems prove its nondegeneracy; in the directed case the joint statement follows also from the conditional Hessians in Section 4. Let $\mathcal{E}_T$ denote only the original physical turn edges, not the hidden unturned resolution wires. Set

$$\begin{aligned} \mathcal{K}(x,t) &= 2J(x) - J(x+t) - J(x-t), \\ a(D) &= -w(D) - \sum_v n_v, \\ \Phi_t(X) &= \Phi(X) + \sum_{e \in \mathcal{E}_T} \mathcal{K}(x_e, t) + \frac{\pi \mathrm{i} a(D)}{2} t^2, \\ S(t) &= \Phi_t(X_*(t)), \qquad X_*(0) = X_0. \end{aligned} \tag{1.4}$$

The analytic implicit function theorem defines this stationary branch for small t. It is an auxiliary local family, not an assertion about the physical invariant with $s = Nt$.

Theorem 1.1 (All orders on translated microscopic charts)**.** *For each source above, every $L \geq 1$, every compact disk $|s| \leq R < 1$, and every fixed compact K in the open upper half-plane,*

$$\log \mathcal{Z}_N(s) = \sum_{r=1}^{L} N^{-r} p_r(s) + O(N^{-L-1}), \qquad \deg p_r \leq r+1. \tag{1.5}$$

The logarithm is the branch tending uniformly to zero. The estimate holds with every prescribed finite number of derivatives on smaller compacts. The coefficients are the same on all these charts. They may be separated by level parity if a lower amplitude requires it; their highest-degree coefficients agree. More precisely,

$$[s^{r+1}] p_r(s) = [t^{r+1}](S(t) - S(0)). \tag{1.6}$$

All coefficient polynomials are even and vanish at zero. In particular

$$p_1(s) = \frac{1}{2} S''(0) s^2, \qquad S''(0) = \pi \mathrm{i} a(D) - 2\pi \sum_{e \in \mathcal{E}_T} \cot(\pi x_e(X_0)). \tag{1.7}$$

The constants can depend on the fixed diagram, powers, certificate, compact, derivative order and expansion order.

A recurrence here means a specified finite operator

$$P(q,M,T)=\sum_{j=0}^{r_P}a_j(q,M)T^j,\qquad TM=qMT,\qquad a_j\in\mathbb{C}(q,M),\tag{1.8}$$

which annihilates the whole reduced function $F_N(M)$ for every sufficiently large integer level. Neither interpolation from finitely many levels nor an equation for the values $F_N(1)$ is this hypothesis. Negative shifts can first be removed by multiplication by a power of T. Clear its fixed rational denominators and divide by the largest common power of $q-1$. Its nonzero specialization is $P_0(M,L)$. No existence claim for a recurrence is needed for the following implication.

Theorem 1.2 (A classical germ forced by the actual recurrence)**.** *For every operator* (1.8) *with the stated all-level property, define*

$$\mathcal{S}(u)=\pi\mathrm{i}\{S(u/(\pi\mathrm{i}))-S(0)\},\qquad L_{\mathrm{an}}(e^u)=-e^{\mathcal{S}'(u)}.$$

Then $P_0(e^u,L_{\mathrm{an}}(e^u))=0$ *as a holomorphic germ. Moreover, writing* $M=e^{\pi\mathrm{i}t}$ *and* $E_e(M)=e^{2\pi\mathrm{i}x_e(X_*(t))}$*,*

$$L_{\mathrm{an}}(M)=-M^{a(D)}\prod_{e\in\mathcal{E}_T}\frac{M^2-E_e(M)}{1-M^2E_e(M)}.\tag{1.9}$$

The selected irreducible Laurent-polynomial factor of this germ therefore divides every such P_0*. This conclusion does not identify a full polynomial, multiplicity or recurrence ideal.*

The analytic assertion is proved without a representation-variety argument. Section 8 states the additional marked geometric input that identifies L_{an} with the preferred longitude, rather than identifying two germs from a common tangent.

The broader comparison of quantum recurrences with peripheral curves is the subject of the AJ conjecture. In its colored-Jones formulation, the conjecture compares the minimal color-shift recurrence at $q=1$ with the A-polynomial, with the coordinate convention and factors depending only on the meridian treated explicitly [10, Section 2.2, Conjecture 2]; see also the factor formulation in [11, Section 1.1]. Here the hypothesis is an operator identity for the whole meridian function at every sufficiently large root-of-unity level. The conclusion selects one analytic germ and, after the separate geometric argument, its preferred-longitude factor. It neither supplies that operator nor identifies the minimal colored-Jones recurrence or the other character components. These references explain the comparison; their proofs are not inputs below.

2 Finite source estimates on the shifted charts

The exact moving finite calculus is imported from [2, Proposition 6.1, Theorem 7.1 and Proposition 8.1]: these are respectively the full crossing polynomial and trace, the original tree contraction, and its scalar ledger. In its fixed Newton basis, with $d=N-1$,

$$e_k(z)=S_k^{-2}\prod_{j<k}(c_j-z),\quad S_k=\prod_{r=1}^{k}2\sin(\pi r/N),\quad c_j=2\cos((2j+1)\pi/N),$$

the quotient is $P_\sigma=\mathsf{T}_N+2\cos 2\pi\sigma$, $\mathsf{T}_N(u+u^{-1})=u^N+u^{-N}$. The metric is $\phi_\sigma(e_ie_j)=\mathbf{1}_{i+j=d}$ with $\phi_\sigma=(-1)^dN^2[z^d]\operatorname{rem}_{P_\sigma}$. An original physical edge, in lower-index convention, has equal indices and coefficient $\gamma_{e,N}(s)H_N(s)/W_k(s)$, where

$$W_k(s)=\prod_{r=1}^{k}\left(1-\frac{\sin^2(\pi s/N)}{\sin^2(\pi r/N)}\right),\qquad \gamma_{e,N}\in\{1,q^{s^2/2},q^{-s^2/2}\}.\tag{2.1}$$

The complete Hermite norm uses all distinct node values and the first derivative at each double node. The odd node -2 is simple. Polynomial equivalence with interval and Chebyshev norms is [4, Lemma 2.1]; quotient products of fixed valence are first reduced, so the assertion does not identify the full matrix algebra with a commutative polynomial algebra.

Lemma 2.1 (Translated compact sets)**.** *For every source class in Section 1, the local selected source, its nonzero leading coefficient, and all fixed exponential error margins hold on a fixed compact $K \subset \{\Im s > 0\}$. Only bounded or polynomial prefactors and starting levels change. The same is true on a finite union of translates $K + j$.*

Proof. For strict Conway sources this is part of [3, Theorems 1.1–1.2]. We give the finite argument also needed for the all-circle classes. Euler's sine product gives

$$W_k(s) = \prod_{\substack{1 \le r \le k \\ a \in \mathbb{Z}}} \left(1 - \frac{s^2}{(r + aN)^2}\right). \tag{2.2}$$

The nonzero integers are distinct. On $K \cup (-K)$ the finitely many small factors are bounded above and away from zero, and the logarithms of all remaining factors are bounded by C_K/m^2. Products of the minima with one and maxima with one give bounds for every subset, uniformly in N, k. Thus W_k, its inverse, H_N and its inverse are uniformly bounded. Positivity in the all-circle estimates is used only for the complete Newton factors; these bounded multipliers do not change their exponential budgets.

For clarity the exact fixed-power interpolation is

$$r^{[n]}_{N,\sigma}(z) = -\frac{1}{N(U - U^{-1})} \sum_{j=0}^{N-1} \lambda_j^n (t_j - t_j^{-1}) \frac{P_\sigma(z)}{z - c_j(\sigma)}, \tag{2.3}$$

where $U = e^{2\pi i\sigma}$, $t_j = q^{-2j-1+2\sigma}$, $c_j(\sigma) = t_j + t_j^{-1}$ and $\lambda_j = (-1)^j q^{j(j+1)-2\sigma j}$. The full power includes κ_σ^n with $\kappa_\sigma = q^{(d+\sigma)^2/2}$. The denominator is bounded away from zero on the nonreal compact. Nodes are in a Chebyshev ellipse of angular height $O_K(1/N)$. The divided difference and its derivative have polynomial bounds there, while the fixed positive or negative powers of the eigenvalues and carrier have bounded modulus. This proves the same full-polynomial bounds as on a disk. Exact Chebyshev folding costs a bounded factor per fold; the number of folds is fixed by the vertex valence.

At a constant opposite-color vertex keep a separate formal σ and use its whole trace on a parameter circle avoiding collisions. Its confluent normals are removable by the finite crossing theorem. The maximum principle for the whole trace minus its finite approximation brings this formal parameter to zero, while the edge parameter s stays independent. This is not an identification of constant and moving orientation types.

The actual trace grid is shifted by $2\pi\sigma/N$. Its exact pole kernel has vertical-side value $iNe^{-Nv}/(1+e^{-Nv})$ after that translation. Thus the endpoint action loss survives the $O_K(1/N)$ displacement. A translated chart $s + j$ may wrap finitely many grid points across $\pm\pi$; retain those entries and their chirps. Their endpoint loss persists, but the chirp itself is not periodic. Paired charge coefficients are

$$(-1)^{d(\ell-1)} U_{\ell-1}(\cos 2\pi\sigma), \qquad |U_{\ell-1}(\cos 2\pi\sigma)| \le \ell e^{2\pi|\Im\sigma|(\ell-1)}. \tag{2.4}$$

The added fixed exponential in ℓ is absorbed by the original $e^{-N\ell y}$ tail at every fixed positive contour height y. The first coefficient stays exactly one. Real/strip estimates in [6, Sections 2–6] and the conditional estimates in [7, Sections 3–7] use precisely these bounded multipliers, polynomial finite norms and geometric tails. All exponential margins therefore persist. This verifies their source hypotheses on K directly; it is not analytic continuation of an asymptotic estimate through resonant real parameters. □

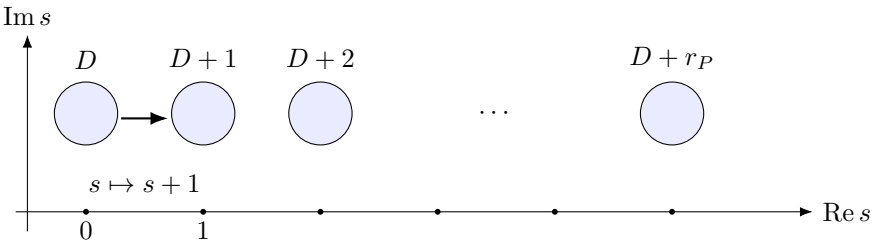


Figure 1: A fixed disk D above the real parameter axis and its translates $D+j$ provide the arguments in an order-r_P recurrence. The real dots are integer resonances; no contour crosses them. The physical meridians are $M = q_N^s$, so every fixed disk still shrinks to $M = 1$ as N grows.

3 Canonical local integrals to every order

The canonical factorial is

$$G_N(z) = \left(\frac{2\pi}{N}\right)^z \Gamma(z+1) \prod_{a\geq 1} \frac{\Gamma(Na+z+1)}{\Gamma(Na-z)(Na)^{2z+1}}. \tag{3.1}$$

We use the complete proof in [5, Lemma 2.1], including its fixed normalization, sine shift and reflection:

$$G_N(k) = S_k, \qquad G_N(z)G_N(N-1-z) = N, \qquad G_N(s)G_N(-s) = H_N(s)^{-1}.$$

On every smaller complex bulk compact it supplies, with all fixed derivatives and every requested finite order,

$$\log G_N(Nx) = NJ(x) + \tfrac{1}{2}\log(2N\sin\pi x) + \sum_{j=1}^{L} N^{-j} b_j(x) + O(N^{-L-1}). \tag{3.2}$$

This is an asymptotic expansion, with no convergence assertion. The summable Gamma remainders in that lemma justify the infinite product before differentiation.

The exact local circle factor and the physical edge insertion are

$$\mathcal{E}_N(x,\tau) = \frac{G_N(d/2+u)G_N(d/2-u)}{G_N(d/2-Nx+u)G_N(d/2-Nx-u)G_N(Nx)^2}, \quad u = \frac{N\tau}{2\pi}, \tag{3.3}$$

$$\mathcal{I}_N(x,s) = \frac{G_N(Nx)^2}{G_N(Nx-s)G_N(Nx+s)}. \tag{3.4}$$

The sine shift proves their equality with $e_k(-2\cos\tau)$ and H_N/W_k at $x = k/N$. In particular the reflection factor in (3.1) puts H_N in the numerator of the edge weight. Hidden complete gates are ratios of the same factorials. Every argument, including $d/2 = N/2 - 1/2$ and every $d-k$ offset, remains exact.

Lemma 3.1 (Uniform local expansion)**.** *On the selected primitive wire contours and the selected local angular contours, the canonical integrals have expansions to every relative inverse-level order. Their coefficients and remainders are holomorphic on the stated smaller parameter compacts. This applies to a conditional edge integral with its parent angle still free, and to a joint strict Conway or summed-circle integral.*

Proof. All factorial arguments lie in fixed bulk complex tubes on these contours. For strict sources this is proved in [3, Section 6]; hidden unturned indices need not be below half. The exceptional half physical edge is bulk and causes no pole. For all-circle sources the edge saddle is in the fixed tube about $1/5 < x < 3/10$, as proved in [6, Section 3] and [7, Section 3]. Substitute (3.2) into the exact ratios. Powers of N, integer offsets and square-root marks give the original leading amplitude and complete inverse-power expansions.

The finite lattice is handled before asymptotic integration. Enclose each primitive coordinate interval by half-grid rectangles and use $\pi N \cot(\pi N z)$. On horizontal sides the difference from its constant value is $O(e^{-2\pi N h})$; on vertical sides it is bounded. Choose the height h before the parameter tube, with the phase gain Ch^2 less than both a fixed fraction of the boundary loss and πh. Thus nonconstant modes and boundary currents have a fixed exponential loss. The constant terms give exactly N^K times the local integral, up to those errors. The primitive lattice and all offsets are the ones in the imported finite source, so no sublattice multiplicity is introduced.

Translate the local contour to its selected critical point. Its real quadratic part is uniformly negative on compact parameter sets; its inverse Hessian and leading amplitude are bounded and nonzero there. To obtain relative order L, Taylor expand phase and amplitude to a sufficiently high finite order, set $X - X_* = N^{-1/2}Y$, and split at $|X - X_*| = N^{-\epsilon}$, with $0 < \epsilon < 1/6$ chosen first. Outside this shrinking region the quadratic bound is $e^{-cN^{1-2\epsilon}}$. Inside it expand the exponential of the cubic and higher terms with its integral remainder. Taking enough Taylor terms makes that remainder at most N^{-L-1} times an integrable polynomial times $e^{-c|Y|^2/2}$. The finitely many Gaussian moments give integer relative powers: odd total moments vanish. Fixed derivatives are obtained by starting with more terms and using a smaller tube.

The absolute Gaussian integral is bounded by a constant times the modulus of its nonzero leading complex integral. Thus a uniform holomorphic relative remainder stays of the same order after integration. The bound may depend on fixed degrees and powers, which is permitted. This proves the claim directly, rather than promoting a first-order estimate to all orders without a remainder argument. Conditional angular and edge integrations use the same proof with the unevaluated parent coordinate as parameter. □

For the strict Conway and summed-circle classes the fixed global source and charge gaps are already below the same joint first charge. Lemmas 2.1–3.1 therefore give their entire source to every relative order. The source error is kept additive until that comparison; it is never compared with a product of separately damped local contributions. The directed case requires one additional normalized induction, which we now give.

4 All orders in the directed subtree contraction

We recall exactly what the directed certificate controls. Put $a_0 = 1/(2\pi)$, $k_0 = 2^{-16}$, $R_0 = 1/8$, $b_0 = 2^{-11}$, $r_0 = 1/16$, $t_0 = 2^{-14}$ and $\chi = 1 - \cos r_0$. Choose a root and, at every nonroot vertex, $0 < c_v < k_0$ and $\delta_v = \min\{b_0/2, c_v\chi/[4(1+c_v)]\}$. Let q_v be one at a nonroot vertex and zero at the root. With children $\mathrm{Ch}(v)$ define

$$a_v = \frac{d_v}{2\pi}, \qquad b_v = \frac{n_v}{2\pi}, \qquad D_v = a_v - \mathrm{i}b_v,$$
$$A_v = \frac{2}{\pi^2}\left(q_v k_0 + \frac{1}{2}\sum_{w\in\mathrm{Ch}(v)} c_w\right), \qquad B_v = 16q_v + |\,\mathrm{Ch}(v)|, \qquad C_v = B_v + \frac{b_v^2}{4A_v}. \tag{4.1}$$

All A_v, B_v are positive because the tree has an edge. Put $\delta_v^{\mathrm{in}} = \min(\{b_0\} \cup \{\delta_w : w \in \mathrm{Ch}(v)\})$,

$$y_v = \min\left\{ \frac{\delta_v^{\mathrm{in}}}{2}, \frac{t_0|D_v|}{4C_v}, \frac{A_v\pi^2}{8(\pi|b_v| + t_0|D_v|)}, \sqrt{\frac{A_v\pi^2}{8B_v}} \right\},$$

$$\beta_v = \frac{210}{361}\frac{a_v}{|D_v|^2}, \qquad \eta_v = \min\left\{ \tfrac{1}{2}t_0|D_v|y_v, \tfrac{1}{2}A_v\pi^2, A_v r_0^2, \tfrac{1}{8}a_v r_0^2 \right\}. \tag{4.2}$$

The criterion is $t_0|D_v| \geq 2$ and $\beta_v < \eta_v/2$ at every vertex, and $\beta_v < (k_0 - c_v)\chi$ at each nonroot vertex. These are the same finite inequalities as in [7].

The full edge polynomial is

$$\mathscr{K}_N(t, u; s) = \sum_{k=0}^{N-1} e_k(-2\cos t)e_k(-2\cos u)\frac{H_N(s)}{W_k(s)}. \tag{4.3}$$

Its trigonometric degree in either angle is at most $N-1$; there is no high-index truncation in this definition. Let $M_{v,N}(u; s)$ denote the exact polynomial sent by a subtree to its parent angle u. It is the actual trace of the parent kernel times the exact child messages, with the original typed scalar. At the root the parent kernel is absent. This is an evaluation order for a fixed contraction, not a new diagram.

The written induction in [7, Sections 5–7] proves, on the parent rectangle $|\Re u| \leq R_0$, $|\Im u| \leq b_0$, a nonzero conditional first-charge saddle. Its local real second derivative is at most $-(19/20)a_v$ and its complex Hessian modulus is at least $(19/20)|D_v|$. All tail, endpoint, truncation and connector errors are below that first charge by a positive margin at least $\eta_v - \beta_v$. Each retained charge $\ell \geq 2$ has additional loss

$$\frac{190}{441}\frac{a_v}{|D_v|^2}(\ell^2 - 1). \tag{4.4}$$

The outgoing real-envelope margin is $(k_0 - c_v)\chi - \beta_v > 0$ when an outgoing edge exists. There is no outgoing requirement at the root. These assertions are the conditional conclusions of that proof, not consequences inferred solely from its final scalar asymptotic. The full kernel, all normal data and the exact child scalars are retained there. Lemma 2.1 verifies these same finite interfaces on the translated parameter compacts. For one fixed finite tree all these finitely many positive margins have positive minimum. The infinite charge tail is one geometric series, not a minimum of infinitely many unrelated gaps.

Proposition 4.1 (Normalized induction to arbitrary order)**.** *Every fixed directed certificate above gives all relative orders of the actual normalized subtree messages on smaller parent rectangles and the meridian compacts of Theorem 1.1. Their phase is stationary elimination of (1.3), with the exact scalar factors restored at the root.*

Proof. Strengthen the induction to

$$\frac{M_{v,N}(u; s)}{M_{v,N}(0; s)} = e^{Nh_v(u)}\frac{\alpha_v(u)}{\alpha_v(0)}\left(1 + \sum_{j=1}^{L} N^{-j}a_{v,j}(u, s) + O(N^{-L-1})\right). \tag{4.5}$$

Here $h_v(0) = 0$, α_v is nonzero, and the coefficients and remainders are holomorphic. The exact denominator is nonzero for all sufficiently large levels by the conditional first-charge estimate. It is never replaced by an estimate for its modulus.

For a terminal vertex, Lemma 3.1 applied to its full kernel and trace proves (4.5); its complement and currents retain the original positive margins. Suppose it holds for all

children. Their finite product has an expansion to every chosen relative order, with a nonzero normalized leading amplitude. On the first-charge local contour use that expansion and the canonical parent-kernel expansion. The conditional Hessian and sine factor are nonzero. Absolute Gaussian domination from Lemma 3.1 preserves the relative remainder, and extra derivatives are obtained by beginning at a higher order. This proves the first-charge expansion and its exact normalization at $u = 0$.

Outside that contour continue to use the exact child polynomials and their established real and strip envelopes. Taylor polynomials for the child expansion are not substituted on a tail. Nonminimal charges are their local contributions plus additive errors already below the first charge; a relative expansion at an exponentially small nonminimal saddle is unnecessary. Inequality (4.4) and the original endpoint and regeneration margins make these errors smaller than any prescribed relative inverse power. The outgoing-envelope step again uses the exact normalized message. It is unaffected by the additional local Taylor coefficients. This closes the induction on the finite tree.

To identify the joint phase, restore the wire variables in each parent kernel. Their Hessian blocks are invertible. At each angular elimination the conditional Hessian is invertible by the displayed bound. The block determinant identity therefore proves that the original joint Hessian is invertible, and back-substitution recovers its selected critical point. No equation for a fixed total action was guessed from the local critical values. Exact factors $M_{v,N}(0; s)$ restore every constant subtracted in the normalized messages. At any finite order, repeated Gaussian integration gives the same moments as the joint local integral: integrating an invertible block Gaussian and then its Schur complement is the usual completion of the square, and differentiation generates all polynomial moments. The uniform absolute bounds already proved justify that finite repeated integration. Thus both the phase and the lower coefficients belong to the original joint source. □

This argument allows constants depending on the fixed tree's size and depth, as an all-orders theorem for one fixed diagram should. It neither strengthens the directed criterion nor applies the summed criterion to a tree which only satisfies local inequalities.

5 The stationary deformation and the polynomial filtration

Set $\delta = 1/N$ and introduce an independent small variable t. Equation (3.2) gives, to every chosen finite order,

$$\log \mathcal{I}_N(x, s) = \delta^{-1}\mathcal{K}(x, t) + B_0(x, t) + \sum_{j=1}^{L} \delta^j B_j(x, t) + O(\delta^{L+1}), \qquad t = \delta s. \tag{5.1}$$

Each B_j is holomorphic, even in t, and zero at $t = 0$. The $\log N$ terms cancel. Explicitly, B_0 is half the same second difference of $\log(2 \sin \pi x)$; the further terms are second differences of the b_j. For a literal argument $x = \ell(X) + \delta c$, expansion of the leading term contributes $c\mathcal{K}_x(\ell, t)$ to B_0 and further holomorphic terms to the lower coefficients. This preserves every d and midpoint offset without changing the leading phase.

In the local first-charge integral replace the physical insertion by the same canonical ratio at $s = Nt$ for small independent t. All its arguments remain bulk. This defines an auxiliary analytic family whose leading phase is exactly (1.4): the last quadratic term is the exact scalar $q^{a(D)s^2/2}$. It defines no global asymptotic statement for a physical parameter growing with N. All global estimates were established for bounded s first. On restricting $t = \delta s$ the auxiliary local family equals the original local integrand.

The joint Hessian of Φ is invertible, by the source theorem in the strict and summed cases and by Proposition 4.1 in the directed case. Thus $X_*(t)$ is analytic and even. Apply the

finite-order Gaussian construction of Lemma 3.1 to the auxiliary local family, uniformly for small t. The nonzero leading amplitude has a fixed local logarithm. It gives holomorphic germs $C_j(t)$ with

$$\log \mathcal{Z}_N(s) \sim N\{S(s/N) - S(0)\} + \sum_{j\geq 0} N^{-j}\{C_j(s/N) - C_j(0)\}. \tag{5.2}$$

The equality means every specified finite truncation with a uniform asymptotic remainder on the charts already proved. For the directed proof the iterated local family has precisely these coefficients: its block stationary elimination and Gaussian moments were identified in Proposition 4.1. Exact child constants are restored, not suppressed in (5.2). In every class the powers of N and all s-independent scalar factors cancel in the ratio at $s = 0$.

Every coefficient in this local family is even in t. Uniqueness of the critical branch preserves that symmetry, as do the exact edge ratios and the scalar. Hence the C_j and S are even. Possible lower-coefficient parity dependence can be carried through this construction separately; its leading phase is the same. Expand the germs in (5.2). A term t^k in $N(S-S(0))$ contributes $N^{1-k}s^k$, while a term t^k in $N^{-j}(C_j - C_j(0))$ contributes $N^{-j-k}s^k$. At order N^{-r} only the first expression can have degree $r+1$; all the others have degree at most r. This proves the filtration and (1.6). It is a coefficient calculation at every finite order, not an inference from the first correction.

All discarded terms have fixed exponential losses on each fixed chart, so they and their fixed derivatives are smaller than every requested inverse power. The coefficients on translated charts are the same because they were constructed from the same canonical local family and scalar. No continuation of an asymptotic estimate through the real resonances is used. Since $\mathcal{Z}_N \to 1$ uniformly on each chart, the logarithm tending to zero is unique for large N. This proves Theorem 1.1.

The first coefficient has no lower-amplitude contribution: $C_0'(0) = 0$ by evenness, and normalization removes the constant. Moreover $\Phi_{Xt}(X_0, 0) = 0$, so stationary differentiation gives

$$S''(0) = \Phi_{tt}(X_0, 0) = \pi \mathrm{i} a(D) - 2\sum_e J''(x_e(X_0)) = \pi \mathrm{i} a(D) - 2\pi \sum_e \cot \pi x_e(X_0).$$

This proves (1.7) without a geometric identification of the resulting complex number.

6 The exact reduced-to-full gauge

The full modified dimension in the normalization of [1, equation (10.11)] is

$$d_N(M) = \mathrm{i}^{N+3} \frac{M - M^{-1}}{M^N - M^{-N}}, \qquad d_N(1) = \frac{\mathrm{i}^{N+3}}{N}. \tag{6.1}$$

The value at one is a limit in M. It satisfies $d_N(q_N^s)/d_N(1) = H_N(s)^{-1}$. Thus (1.1) is the full state divided by its value at one. Write $f(M) = M - M^{-1}$. Since $q_N^N = -1$, for every integer j,

$$\frac{d_N(q_N^j M)}{d_N(M)} = (-1)^j \frac{f(q_N^j M)}{f(M)}, \qquad \frac{H_N(s+j)}{H_N(s)} = (-1)^j \frac{f(M)}{f(q_N^j M)}. \tag{6.2}$$

These are exact identities, including the meridian divisor.

Let P be as in (1.8), after clearing its original rational denominators and removing its common $q-1$ valuation. Substituting $F_N(q_N^s) = K_N H_N(s) \mathcal{Z}_N(s)$ in its actual equation

shows that $\mathcal{Z}$ satisfies the fixed rational Ore operator

$$\widetilde{P} = \sum_{j=0}^{r_P} a_j(q, M)(-1)^j \frac{f(M)}{f(q^j M)} T^j. \tag{6.3}$$

Multiplication on the left by $\prod_{j=1}^{r_P} f(q^j M)$ clears these added denominators. The first nonzero classical symbol is

$$f(M)^{r_P} P_0(M, -L). \tag{6.4}$$

It is nonzero as a Laurent polynomial. No meridian factor is removed by a convention about the reduced invariant.

Choose an open disk D in the upper half-plane and apply Theorem 1.1 to slightly enlarged compact disks covering $D, D+1, \ldots, D+r_P$. The exact cleared equation holds there for every sufficiently large level, and $\mathcal{Z}$ is nonzero. Divide by $\mathcal{Z}_N(s)$ and compare successive powers of N^{-1}. The uniqueness of asymptotic coefficients follows by multiplying by the first putative nonzero power and taking the limit; it holds along either parity sequence. Every resulting coefficient vanishes on D. Since the coefficient is a polynomial in s, obtained by the finite shifts and the Taylor series of Laurent polynomial coefficients at $(q, M) = (1, 1)$, it vanishes identically. This step uses expansions at all the shifted arguments of the actual equation. Complete values or a chart restricted to $|s| < 1$ would not justify an operator of arbitrary fixed order.

7 Recovering the classical characteristic

Put $h = \log q$ and $u = \log M$, so the physical substitution is $u = hs$ and $h = \pi\mathrm{i}/N$. The filtration gives a unique regrading of the formal logarithm as

$$h^{-1}\mathcal{S}(u) + \sum_{a \geq 0} h^a A_a(u), \qquad \mathcal{S}(u) = \pi\mathrm{i}\{S(u/(\pi\mathrm{i})) - S(0)\}. \tag{7.1}$$

Indeed a monomial $h^r s^k$ becomes $h^{r-k}u^k$; $k \leq r+1$ permits just the one negative power h^{-1}. For each exponent of h its coefficient is a formal series in u. Its leading part is fixed by (1.6). Under the jth shift, u changes to $u + jh$. The difference of (7.1) has no negative power of h, because $\mathcal{S}(u+jh) - \mathcal{S}(u)$ is divisible by h. Its constant coefficient is $j\mathcal{S}'(u)$. The lower A_a contribute no constant coefficient. Consequently the formal shift ratio belongs to $\mathbb{C}[[h, u]]$ and has constant h-coefficient $e^{j\mathcal{S}'(u)}$. Its exponential is well defined because $\mathcal{S}'(0) = 0$.

We give the needed injectivity explicitly. For $B(h, u) = \sum_{a,b \geq 0} B_{a,b} h^a u^b$, substitution $u = hs$ gives

$$[h^n s^b]B(h, hs) = \begin{cases} B_{n-b,b}, & b \leq n, \\ 0, & b > n. \end{cases} \qquad B_{a,b} = [h^{a+b}s^b]B(h, hs). \tag{7.2}$$

At each n there are finitely many terms, so this is a well-defined ring map into $\mathbb{C}[s][[h]]$, and the second equality proves injectivity. Agreement through order h^L recovers exactly the triangle $a + b \leq L$, not all coefficients. For example h^{L+1} is invisible at that finite order. We need, and have proved, every finite asymptotic order.

The formal operator equation from Section 6 therefore holds before substitution. Take its constant h-coefficient and use (6.4) to obtain

$$f(e^u)^{r_P} P_0(e^u, -e^{\mathcal{S}'(u)}) = 0.$$

The ring of formal series is a domain and $f(e^u)$ is nonzero. Cancel it in this identity, without altering the operator, to get

$$P_0(e^u, -e^{\mathcal{S}'(u)}) = 0. \tag{7.3}$$

The germ S is analytic by its local implicit construction, so this formal equality is also an equality of holomorphic germs.

Stationary differentiation of (1.4) gives its exact rational form. With $M = e^{\pi i t}$ and $E = e^{2\pi i x}$,

$$e^{\mathcal{K}_x(x,t)} = \frac{(1-E)^2}{(1-EM^2)(1-E/M^2)}, \qquad e^{\mathcal{K}_t(x,t)} = \frac{M^2-E}{1-M^2E}. \tag{7.4}$$

These follow by substituting $2\sin\pi z$ in $\mathcal{K}_x = 2J'(x) - J'(x+t) - J'(x-t)$ and $\mathcal{K}_t = J'(x-t) - J'(x+t)$; their logarithms are zero at $t = 0$. The derivative of the stationary value differentiates only the explicit parameter. Thus $\mathcal{S}'(u) = S'(u/(\pi i))$ and (1.9) follows. All denominators are units near the selected point since the original filled bulk labels have $E_e \neq 1$; a half label $E_e = -1$ is harmless. This proves the analytic part of Theorem 1.2.

For any one specified operator in that theorem, evaluation on this germ is a homomorphism $\mathbb{C}[M^{\pm1}, L^{\pm1}] \to \mathbb{C}\{u\}$ with prime kernel. The kernel is nonzero by (7.3). It contains no nonzero Laurent polynomial in M alone because $M = e^u$ is nonconstant. Its height is therefore one: a height-two prime in this two-dimensional Laurent polynomial ring would give an algebraic constant value of M. This ring is a unique factorization domain, so its height-one prime is generated by one irreducible Laurent polynomial, up to a monomial unit. Every P_0 in the theorem belongs to it. Over a smaller coefficient field the contraction of this prime can include conjugate factors; no claim that an individual complex factor is defined over that field is needed. The conclusion concerns the selected germ and its factor, not all components or their multiplicities.

8 Identifying the preferred longitude

The preceding theorems construct an analytic characteristic from an actual source and a specified recurrence. Its geometric identification uses two separately proved classical statements. For all-circle trees, [8, Theorem 1.2 and Corollary 7.1] construct the full original-group filled germ at its positive hyperbolic endpoint, with the M^{-1} flag and primitive logarithms. They give uniqueness after the actual endpoint and those marks have been identified. For strict sources with bare junctions, [9, Theorem 1.3] identifies the stationary branch of precisely (1.4), retaining the original physical edges and all bare junctions. Its proof uses classical zero-circle insertion and elimination; it includes the half-label fold and makes no quantum asymptotic assumption.

We spell out the all-circle interface to avoid replacing a group identification by an equality of second derivatives. Put $Q_v = e^{i\tau_v}$ and $E_e = e^{2\pi i x_e}$. Differentiating (1.2) and using (7.4), the wire stationary equation is exactly

$$(E_e + Q_v)(E_e + Q_v^{-1})(E_e + Q_w)(E_e + Q_w^{-1}) = (E_e - 1)^2(1 - E_eM^2)(1 - E_e/M^2). \tag{8.1}$$

The local source log satisfies

$$-2\pi b_\tau(x,\tau) = \log\frac{Q+E}{1+QE}. \tag{8.2}$$

To check its branch as well as its derivative, differentiate both sides in x: both derivatives are $2\pi\sin\tau/(\cos 2\pi x + \cos\tau)$; both sides vanish at $x = 0$. Continue from the same complete chart. The original angle stationarity therefore gives

$$\sum_{e\ni v} \log\frac{Q_v + E_e}{1 + Q_vE_e} - n_v \log Q_v = -2\pi i, \qquad n_v = \epsilon_v - 2p_v. \tag{8.3}$$

These are exactly $U_v + p_v V_v = -2\pi\mathrm{i}$, with $U_v = \sum \log((Q_v + E_e)/(1 + Q_v E_e)) - \epsilon_v \log Q_v$ and $V_v = 2 \log Q_v$. The linear first-charge term in (1.3) fixed the right side before exponentiation.

At $t = 0$ the selected source point is the positive smooth endpoint identified by the original summed or directed source theorem; [7, Section 8] includes its backwards restoration from conditional saddles. The summed criterion gives $\sqrt{d_v^2 + n_v^2} \geq \sqrt{2^{25} d_v} > 630$ at every vertex, and the directed criterion gives $\sqrt{d_v^2 + n_v^2} = 2\pi |D_v| \geq 2\pi\, 2^{15} > 630$. Thus the marked geometric theorem applies also to high-degree vertices with zero twist. Its full-group reconstruction uses (8.1) and the original flags, not just traces. At that endpoint $Q_v^2 \neq 1$ for each filled disk. It persists on a neighborhood, so a slope word commuting with its semisimple longitude and having eigenvalue ratio one is projectively identity. Consequently the reconstructed branch is a branch of the actual filled group. The endpoint, flag and primitive logarithmic conditions now meet the hypotheses of [8, Corollary 7.1], which identifies the branch.

For the geometric conclusion in the strict class, enlarge its fixed power threshold to the maximum of the analytic threshold in [3, Theorem 1.2] and the classical positive-filling and longitude threshold in [9, Theorem 1.3]. Both are constants of the same fixed original diagram, so this preserves independent signed powers and imposes no ratio condition. The analytic theorems above themselves require only their analytic source threshold. No inequality between the two thresholds is being assumed.

For a strict source beyond this common threshold the classical action in [9, Theorem 1.3] is the sum of the same m and b functions, with their absolute primitives, the same chirp and the same first-charge term. Hidden unturned wires can be retained before its stationary elimination. Its original physical-edge set is $\mathcal{E}_T$. The nondegenerate source branch proved here therefore equals that classical branch by the local implicit uniqueness theorem; no individual half-edge elimination is used at a folded label.

Corollary 8.1 (The selected geometric factor)**.** *For every fixed source in Section 1, with the strict-class threshold enlarged to the common threshold just specified, the germ L_{an} is its preferred-longitude eigenvalue on the M line, equivalently the inverse value on the source M^{-1} line. It has lift $L_{\mathrm{an}}(1) = -1$. Every specified actual all-level annihilator in Theorem 1.2 contains this selected geometric factor in its first nonzero classical symbol. If $\tau_K = \frac{d}{du} \log[-L_{\mathrm{an}}(e^u)]\big|_{u=0}$, then uniformly on the stated charts,*

$$\frac{F_N(q_N^s)}{K_N H_N(s)} = 1 + \frac{\pi\mathrm{i}\tau_K s^2}{2N} + O(N^{-2}).$$

Proof. The branch identification just established permits the marked longitude formulas [9, Theorems 1.2–1.3]. Their rational product is (1.9), including its integer exponent $a(D)$ and negative determinant-one sign. Those constants are fixed there by peripheral and framing calculations, not by the quantum tangent. Hence $\tau_K = \mathcal{S}''(0) = S''(0)/(\pi\mathrm{i})$. Theorem 1.2 proves the factor statement, and (1.7) gives the expansion. □

9 Finite reproduction and boundaries

The accompanying `check.py` checks the finite rational reduced/full gauge for several operator orders, the rational sine ratios in (7.4), and coefficient recovery and finite-jet truncation for sparse formal polynomials. These are algebraic checks of displayed identities. They do not establish analytic remainders, global source gaps, existence of a recurrence, or a geometric branch by testing levels. Those are the written proofs and explicitly named companion inputs above.

The expansion is microscopic on each fixed compact in s. The auxiliary t family organizes its local coefficients and creates an analytic germ, but does not supply a global asymptotic for fixed $M \neq 1$. The recurrence hypothesis is one fixed finite operator valid for the whole function at all sufficiently large levels. The conclusion selects one germ and one irreducible factor; it does not assert equality with a full A-polynomial or an operator ideal. None of these distinctions alters the original all-level and both-parity source conclusions.

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